

Powering the Always-On Economy

A Plain-Language Guide to Generation Mix Decisions for Continuous Load

Companion to: Generation Mix Optimization Under Competing Objectives

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Introduction: A New Kind of Customer

Something new is being plugged into the North American power grid. Not a factory that runs a day shift. Not a shopping mall that dims its lights at midnight. Not a neighborhood that draws more power in the evening and less at three in the morning. What is being plugged in is a hyperscale data center — a building the size of several city blocks, drawing between 100 and 500 megawatts of electricity continuously, at the same level, every hour of every day of the year.

There are hundreds of these facilities either operating or in development across North America. More are being announced every month. The electricity they collectively require is not a rounding error in the national generation picture — it is, by NERC’s own projections, the dominant driver of new load growth across multiple interconnections. ERCOT alone is processing queue requests from data center campuses that would add tens of gigawatts of new demand. Every major transmission region is working through a version of the same problem.

The investment community has recognized the opportunity. Venture capital, private equity, and infrastructure funds are moving capital into data center development at a pace the industry has not previously seen. The bottleneck they are now hitting is not land, not fiber, not cooling technology. It is electricity — reliable, continuous, affordable, and increasingly, low-carbon electricity at a scale that the existing grid was not designed to deliver on demand.

This is, at its core, an investment problem. And, as with most investment problems, it is being analyzed with tools designed for a different question.

This document explains what those tools miss, what a better framework looks like, and what it reveals about the generation choices now facing investors, utilities, and policymakers. The mathematical derivations underlying these conclusions are developed in full in the companion technical white paper, “Generation Mix Optimization Under Competing Objectives: A Parametric Framework for Climate and Resilience Tradeoffs” (Roxey, 2026). This document presents the same argument without the mathematics, for readers who need the conclusions and the reasoning but not the derivations.

Part One: Why the Standard Scorecard Is Wrong

When investors and utilities evaluate generation options, they first reach for a number called the levelized cost of energy (LCOE). It is the all-in cost of building, financing, fueling, operating, and eventually decommissioning a power plant, divided by the total electricity it produces over its lifetime. The result is a price per megawatt-hour that allows different generation technologies to be compared on a common basis.

LCOE is not wrong. It is, however, an answer to a question that the data center investor is not asking.

The data center investor is not asking: “What is the average cost of electricity over a plant’s lifetime under normal conditions?” The data center investor is asking: “What is the cost of electricity that shows up continuously, meets my tenants’ carbon commitments, and keeps showing up when something goes wrong?” These are different questions. LCOE answers the first. It has almost nothing to say about the second.

The gap between those two questions is where the investment errors accumulate. Three things LCOE misses matter enormously for continuous load service.

There is a better metric for this decision. Call it the Resilience-Adjusted Delivered Cost — RADC. It asks a single question: what does it actually cost to deliver continuous, low-carbon, resilient electricity over the life of a twenty-year contract? That question includes three things LCOE ignores: the cost of keeping variable sources running continuously, the carbon commitments the tenants have already made, and the risk that power stops arriving when something goes wrong. RADC is the number a data center investor needs. LCOE is the number that is easy to calculate. They are not the same number, and confusing them is expensive.

The First Miss: Ready Is Not the Same as Running

Every generation source has two different measures of its output performance. The first is availability — the fraction of time the plant is mechanically capable of producing power. The second is the capacity factor — the fraction of the theoretical maximum output it actually delivers, averaged over the year.

For some sources, these two numbers are close together. A nuclear plant is available about 90% of the time and runs at about 92% of its maximum output when available. The two numbers tell roughly the same story.

For others, the gap is enormous. A utility-scale solar farm is available — mechanically ready to produce — about ninety-nine percent of the time. But it actually delivers only about 25% of its theoretical maximum output because the sun does not shine at night and is weaker in winter. The plant is “ready” almost all the time. It is “running” only a quarter of the time.

For a data center that draws the same load at two in the morning as it does at two in the afternoon, the distinction is not academic. The generation serving that load must be running, not merely ready. A source that is ready ninety-nine percent of the time but runs only twenty-five percent of the time cannot serve a continuous load without something else filling the gap during the seventy-five percent of the time it is not producing. That something else costs money — and that cost does not appear in the LCOE figure for the solar farm.

The product of availability and capacity factor — call it productive utilization — is what the data center investor actually needs to know. Nuclear’s productive utilization is eighty-three percent.

Utility solar is twenty-five percent. Utility wind's is thirty-four percent. Gas combined-cycle's is forty-eight percent. These numbers, not the LCOE figures, determine how much of the time the source is actually delivering the continuous power the data center requires.

The Second Miss: Carbon Commitments Are Binding

The companies occupying hyperscale data centers — cloud providers, artificial intelligence infrastructure firms, and financial institutions — have publicly committed to net-zero electricity consumption. These are not aspirational statements. They are contractual obligations embedded in lease agreements, regulatory filings, and investor disclosures. A generation that does not meet this customer class's needs is not a viable option, regardless of cost.

Natural gas combined-cycle generation — currently the fastest and cheapest new-build option available — produces approximately 490 grams of carbon dioxide per kilowatt-hour on a lifecycle basis. Nuclear produces twelve. Wind produces eleven. The gap is not a rounding difference. A data center powered primarily by gas is incompatible with net-zero electricity commitments. Full stop.

LCOE does not capture this. A gas plant with an LCOE of thirty-nine dollars per megawatt-hour looks far cheaper than a nuclear plant at eighty-nine dollars. It is, in fact, unusable for a significant fraction of the data center market.

The Third Miss: Resilience Has a Price Tag

The third thing LCOE misses is the one least visible in current market discussions and, increasingly, the one with the largest potential consequences.

Generation assets are physical infrastructure. They can be damaged. Their fuel supply can be interrupted. Their most critical components can take a year or more to replace. These are not hypothetical risks — they are risks that infrastructure targeting in active conflict zones has demonstrated repeatedly, and that critical infrastructure planners in stable regions are now taking seriously in their long-term assessments.

A generation source whose fuel supply can be interrupted within hours — natural gas, which depends on continuous pipeline delivery — presents a fundamentally different risk profile from one that carries months of fuel on site. A source whose most critical component, if destroyed, would take 18 to 24 months to replace presents a fundamentally different recovery profile from one that can be restored in weeks. These differences are real, quantifiable, and absent from any LCOE figure.

Think of it this way. Two pickup trucks have the same sticker price. One has a two-gallon tank and a fuel door that sticks. The other has a forty-gallon tank and a locking cap. LCOE is the sticker price. The fuel tank and the locking cap are resilient. You would not buy a

vehicle without knowing the tank size. You should not buy a generation without knowing the fuel independence duration.

Part Two: The Real Comparison

Once the three misses are on the table, the comparison between generation options changes substantially. Let's work through what the real comparison looks like for each of the four most plausible new-build options: nuclear small modular reactors, gas combined-cycle, utility wind with storage, and utility solar with storage.

The Firmed Cost: What Continuous Delivery Actually Costs

A source that does not run continuously cannot serve a continuous load without backup. The backup costs money. The fair comparison for a continuous load application is therefore not the LCOE of the source alone — it is the LCOE of the source plus the cost of the storage or backup generation required to make its output continuous.

Using current battery storage costs and the productive utilization figures above, the “firmed” cost of each option — the true cost of continuously delivered electricity — works out as follows:

Nuclear requires very little firming because it is already running ninety-two percent of the time. Its firmed cost is approximately one hundred dollars per megawatt-hour.

Gas combined-cycle requires moderate firming. Its firmed cost is approximately ninety-nine dollars per megawatt-hour — nearly identical to nuclear on this measure alone.

Wind with storage requires substantial firming because wind runs only 35% of the time. Its firmed cost is approximately one hundred and twenty-four dollars per megawatt-hour.

Solar with storage requires the most firming because solar is only running twenty-five percent of the time. Its firmed cost is approximately one hundred and twenty-nine dollars per megawatt-hour.

The headline cost advantage of wind and solar over nuclear — roughly sixty to sixty-five dollars per megawatt-hour on a raw LCOE basis — disappears entirely when the cost of making their output continuous is included. For the specific problem of serving a data center that never sleeps, firmed wind and firmed solar are more expensive than nuclear, not cheaper.

Two important qualifications apply to these firmed cost figures. First, a variable source serving continuous load requires two things simultaneously, not one: additional nameplate capacity to generate enough total energy during the hours it is producing, and storage to hold that surplus energy for the hours when it is not producing. These are additive capital requirements. Neither

alone is sufficient. The firmed cost figures above capture both, but it is important to understand that building four times the solar nameplate capacity without adequate storage solves nothing — the surplus daytime energy is simply wasted, and the overnight gap remains.

Second, the firmed cost estimates are conservative in the direction that favors wind and solar, and for different reasons in each case. For solar, the problem is structural: solar produces nothing for a minimum of ten to twelve hours every night, and longer in winter. Four hours of battery storage gets a data center to midnight — it does not get it to sunrise. Truly firming solar to continuous output requires twelve to sixteen hours of storage at a minimum, substantially more in northern latitudes or winter months. For wind, the problem is duration: four hours of storage handles short lulls, but wind can be calm for multiple consecutive days across wide geographic areas. Bridging those multi-day events requires days of stored energy, not hours. In both cases, the actual firmed cost of continuous delivery is materially higher than the figures stated above.

Capital Recovery: How Long Is Your Money at Risk

The second dimension of the real comparison is how long it takes to get invested capital back. This matters to every investor, but it matters especially in infrastructure, where capital is tied up for decades and the range of things that can go wrong over that period is wide.

Capital recovery is simple in concept. Capital only earns its keep when the plant is running. A plant running at eighty-three percent of its capacity is earning its keep nearly all the time. A plant running at twenty-five percent is sitting idle three hours out of every four, and idle capital does not pay back. In utility finance, this logic has a formal name: the Allowance for Funds Used During Construction, or AFUDC — the accounting construct that makes productive utilization, not installed cost, the binding variable in capital recovery. The same logic applies here.

This arithmetic has a practical consequence that standard analysis misses. The apparent cost advantage of solar and wind on a dollars-per-kilowatt-installed basis evaporates when you account for how long their capital is at risk before it comes back. A solar farm at fifteen hundred dollars per kilowatt that runs at twenty-five percent utilization has its capital at risk for roughly four times as long as a nuclear plant at seventy-five hundred dollars per kilowatt running at eighty-three percent utilization — and over that extended exposure period, the solar investor faces four times as many years of market, regulatory, and physical risk.

That does not make solar a bad investment in every context. It makes solar a poorly matched investment for the specific problem of serving a continuous baseload load, where the premium on reliable, continuous output is highest and the penalty for extended capital recovery periods is steepest.

The Fuel Question: How Long Can It Keep Running

The third dimension is fuel independence — how long a source can keep running if its external fuel supply is cut off.

Nuclear carries enough fuel on site, loaded into the reactor core at the beginning of each operating cycle, to run for approximately two hundred and seventy days without any external fuel delivery. If every road, rail line, and pipeline serving the plant were severed tomorrow, the plant would continue producing electricity for the better part of a year.

A gas combined-cycle plant has approximately twelve hours of fuel in the pipeline lateral serving it — the branch line connecting the plant to the regional gas transmission system, specifically the gas that is already in that pipe between the plant and the nearest compressor station. If gas delivery stops, the plant goes dark within hours.

Coal plants fall between these extremes. Most maintain on-site stockpiles of thirty to ninety days, depending on the facility and the rail or barge supply arrangements. The stockpile is a countdown clock, not a permanent solution, but it buys substantial time.

Wind and solar cannot have their fuel cut off because their fuel is the wind and the sun. But they can have their equipment destroyed, and when that happens, the recovery timeline — the time required to replace the damaged components and restore service — becomes the relevant resilience variable.

For the data center investor signing a twenty-year power purchase agreement, fuel independence duration is a direct measure of force majeure risk. A contract backed by gas generation carries the risk that a pipeline disruption — whether from physical damage, weather, regulatory action, or supply shortage — could void the agreement during exactly the conditions when the data center operator needs power most. A contract backed by nuclear generation carries no equivalent fuel supply risk.

The Equipment Question: How Long to Recover

Every generation source that connects to the high-voltage transmission system does so through a large power transformer — a custom-manufactured piece of equipment that steps the generator's output voltage up to transmission level. These transformers are not off-the-shelf items. They are designed and built to order, one at a time, for specific sites and voltage classes. Under normal commercial conditions, the lead time from order to delivery is twelve to twenty-four months. There is no meaningful strategic reserve of these components in North America.

This means that if a large power transformer is destroyed — by physical attack, by severe weather, by any cause — the facility it serves is offline for the better part of two years while a replacement is manufactured, shipped on specialized heavy transport, and installed. The turbine, the fuel supply, the control room, and the operators can all be perfectly intact. Without the transformer, none of it matters.

Open-source analysis of infrastructure targeting in active conflict zones has documented a specific pattern: adversaries who understand this vulnerability target transformers deliberately, wait for replacements to be installed, and then destroy the replacements. The strategic logic is

straightforward — the replacement inventory is finite, and each replacement cycle depletes it further. This is not a theoretical risk. It is a documented operational practice.

Large hydro facilities face an even longer recovery timeline for their most critical components — the turbine runners are custom-cast for specific sites and can take two to three years to replace. This is the one dimension on which large hydro, otherwise an attractive generation option, carries a more severe risk profile than nuclear or thermal plants.

A Fifth Consideration: Fuel Cells as a Resilience Layer

A technically informed reader working through the comparison above will notice that one energy conversion technology has been absent from the discussion: the fuel cell. The omission is intentional but not dismissive. Fuel cells do not belong in the primary generation comparison because they are not a generation source in the same sense as nuclear, gas, wind, or solar. A fuel cell is a conversion device — it converts a stored fuel, typically hydrogen or natural gas, into electricity through an electrochemical reaction rather than combustion. The critical question is always where the hydrogen comes from, and on that question, the analysis forks sharply.

Where fuel cells become genuinely interesting for data center applications is as a resilience layer — an onsite backup and ride-through capability that addresses some of the vulnerabilities the primary comparison exposes. Microsoft has conducted extended operational testing of hydrogen fuel cells as a direct replacement for diesel emergency generators at data center scale, with results that demonstrate multi-day ride-through capability at megawatt scale. The fuel independence duration metric that figures so prominently in the primary comparison is exactly where hydrogen fuel cells can excel: hydrogen can be produced and stored onsite in quantities that provide days or weeks of independence from the external grid, far beyond the twelve-hour gas pipeline dependency that makes gas-fired generation structurally vulnerable. If the hydrogen is produced from nuclear electrolysis — using surplus nuclear output during off-peak periods to run electrolyzers — the lifecycle carbon of the stored fuel is very low, consistent with the net-zero commitments the primary generation must satisfy.

The honest complications are three. First, if the hydrogen is produced from natural gas reforming — grey hydrogen, the current commercial standard, the carbon problem reappears upstream. The electrochemical conversion in the fuel cell produces no direct emissions, but the production of the fuel carries a carbon burden that the net-zero accounting must absorb. Green hydrogen from electrolysis solves this problem, but the cost of green hydrogen at the scale a data center requires remains substantially higher than grey hydrogen under current market conditions. Second, the production, compression, and onsite storage of hydrogen at data center scale is an engineering challenge that is being actively worked on, but is not commercially solved at the volumes that would make it a standard deployment option. Third, fuel cells do not provide the synchronous inertia that the grid stability analysis in Part Five identifies as a critical and declining resource. A fuel cell connects to the grid through power electronics and offers no rotational inertia

contribution. As a resilience layer, it helps the data center ride through grid events; it does not help the grid itself survive them.

The right place for fuel cells in the data center generation architecture is therefore a second layer, not the first. The primary generation requirement — continuous, grid-connected, low-carbon, synchronous — is best satisfied by nuclear for the reasons the scorecard in Part Three demonstrates. Fuel cells, backed by green hydrogen produced from that nuclear output and stored onsite, then provide the resilience layer that the scorecard’s fuel independence and equipment recovery rows are measuring. Together, the two technologies address both the primary generation requirement and the on-site resilience requirement without compromise on either carbon or continuity. Separately, each leaves a gap that the other fills. An investor or operator evaluating the architecture should think of nuclear and hydrogen fuel cell storage not as competing options in the same column of the scorecard, but as complementary layers in a two-tier resilience design.

Part Three: The Scorecard Redrawn

With the three misses corrected, the comparison between new-build generation options for continuous data center load looks very different from the LCOE table that most investment analyses lead with. The table below presents the comparison across the metrics that actually matter for this application.

Table: What the Real Comparison Shows

What matters	Nuclear (SMR)	Gas (CCGT)	Wind + Storage	Solar + Storage
Does it run continuously?	✓ Yes — 92% of the time	✓ Yes — when gas flows	⚠ Only with storage	⚠ Only with storage
True cost for continuous delivery	~\$100/MWh	~\$99/MWh	~\$124/MWh	~\$129/MWh
Carbon intensity	✓ Very low (12 g/kWh)	✗ High (490 g/kWh)	✓ Very low (11 g/kWh)	✓ Low (48 g/kWh)
How long can it run if fuel is cut off?	✓ 270 days	✗ 12 hours	✓ Fuel cannot be cut off	✓ Fuel cannot be cut off
How long to recover if key equipment is destroyed?	18–24 months	18–24 months	6–12 months	3–6 months
Capital recovery speed	✓ Fastest among low-carbon	✓ Fast	⚠ Slow	✗ Slowest
Passes all tests?	✓ Yes	✗ Fails carbon	⚠ Fails cost	⚠ Fails cost

A note on the table above: fuel cells do not appear as a column because they are not a primary grid-scale generation source. They belong in the architecture as a resilience layer beneath the primary generation — onsite hydrogen storage backed by green hydrogen production, providing multi-day ride-through capability that complements rather than competes with the sources compared here. The relationship between nuclear generation and hydrogen fuel cell storage is addressed in Part Two.

The table tells a clear story. Gas combined-cycle is nearly competitive on firmed cost and capital recovery speed — but it fails the carbon test decisively, and its twelve-hour fuel independence duration makes it structurally vulnerable to supply interruption. It is the right answer for an investor who needs power quickly and is willing to accept both the carbon liability and the supply chain risk. It is the wrong answer for a twenty-year contract serving carbon-committed tenants.

Wind and solar with storage satisfy the carbon test and are immune to fuel supply denial — but they fail the firmed cost test for continuous load. At one hundred and twenty-four to one hundred and twenty-nine dollars per megawatt-hour for continuously delivered electricity, they are twenty-five to thirty percent more expensive than nuclear for the specific application of always-on load. They also carry the longest capital recovery timeline of any option, because their productive utilization rates mean capital is at risk for an extended period before it is returned.

Nuclear is the only option that passes all five tests simultaneously. It is not the cheapest option on any single metric taken in isolation. It is the most cost-effective option when the full bundle of properties the data center application requires is priced correctly.

The right question is not “which source is cheapest?” It is “which source delivers everything this load requires at the lowest total cost?” When the question is framed correctly, the answer is nuclear. Not as a matter of preference — as a matter of arithmetic.

The upfront capital cost of nuclear — the objection most commonly raised against it in investment discussions — is real and should not be dismissed. Current estimates for small modular reactors range from five thousand to ten thousand dollars per kilowatt of installed capacity, compared to one thousand to fifteen hundred dollars per kilowatt for gas and solar. The gap is large.

But the relevant comparison is not dollars per kilowatt installed. It is dollars per kilowatt of continuously delivered, low-carbon, resilient output. When the comparison is made on that basis — accounting for the productive utilization rates that determine how much of the installed capacity is actually doing useful work, the storage costs required to firm variable sources to a continuous profile, and the operating life over which capital is recovered — nuclear’s apparent cost disadvantage shrinks substantially. For facilities licensed to operate for sixty to eighty years

against solar and wind plants designed for twenty-five to thirty, the present value arithmetic changes further in nuclear's favor.

The current U.S. policy environment also provides financing instruments specifically designed to address nuclear's upfront capital challenge: Department of Energy loan guarantees, production tax credits under the Inflation Reduction Act, and investment tax credits for new nuclear construction. Combined, these can reduce the effective cost of capital by several hundred basis points — a reduction that, at nuclear's productive utilization rate and operating life, translates into a material improvement in the investment return. The investor who reaches for these instruments after concluding from the firmed cost comparison that nuclear is the right tool is having a very different financing conversation than the investor who dismisses nuclear based on the raw LCOE figure before the comparison is made correctly.

Part Four: What the Regulators Have Not Done Yet

The investment case for nuclear in the data center context does not depend on regulatory change. The arithmetic works with the rules as they currently stand. But investors should understand three gaps in the current regulatory framework that affect their risk exposure, because those gaps mean that the grid on which their generation assets operate is less protected than they may assume.

The Reserve Margin That Does Not Measure What It Claims

North American grid reliability is governed in large part by planning reserve margins — requirements that utilities maintain installed generation capacity above their peak demand forecast by a specified percentage. The idea is that if some generators are unavailable during a peak demand event, there is enough extra capacity to cover the gap.

Reserve margins are calculated on installed nameplate capacity, not on productive utilization. A region with a fifteen percent reserve margin calculated on nameplate capacity may have far less than a fifteen percent reserve on a productive utilization basis, if the reserve capacity is disproportionately composed of sources that run at twenty-five or thirty-five percent utilization. For a grid adding large amounts of always-on data center load — which is baseload demand, not peak demand — a nameplate-based reserve margin systematically understates the generation gap.

No current North American reliability standard requires reserve margins to be calculated on a productive-utilization-adjusted basis. This is a gap that will eventually be addressed because the arithmetic of the data center demand surge will force the issue. Investors should understand that the reserve margin number they see in NERC assessments is a nameplate figure, not a continuous-delivery figure, and the two can diverge significantly for a grid in carbon transition.

The Fuel Independence Standard That Does Not Exist

No current North American reliability standard establishes a minimum fuel independence duration for generation serving critical load. A generation fleet composed entirely of gas-fired plants — with twelve hours of on-site fuel independence — satisfies all current adequacy standards while being structurally vulnerable to any supply chain disruption that lasts longer than half a day.

This is not a hypothetical concern. Extreme weather events, pipeline failures, and — increasingly — deliberate infrastructure disruption have all produced gas supply interruptions lasting days or weeks. The generation fleet has no mandatory requirement to carry fuel that would survive such events.

For data center investors, the implication is direct: if the generation serving their facility is gas-fired, the power purchase agreement is only as secure as the gas lateral — the branch serving that facility — that connects the plant to the regional pipeline. That lateral is not subject to the same resilience requirements as the generation facility itself. The exposure is real and currently unmitigated by regulation.

NERC has the authority to address this through its existing standards process. A minimum fuel independence duration requirement for generation serving critical load would not require new legislation — it would require a standards revision and the political will to impose a genuine obligation on an industry that currently has none. That revision has not happened. It should be on the horizon for any investor with a long-term perspective on this market.

The Transformer Reserve That Is Not There

The large power transformers that connect generation facilities to the high-voltage transmission system have twelve-to-twenty-four-month manufacturing lead times and no meaningful strategic reserve in North America. This is not a new finding — it has been identified in government assessments, industry reports, and academic analyses for over a decade. It has not produced a mandatory reserve standard.

Current NERC standards under CIP-014 address the physical security of critical transmission facilities. They do not establish recovery timeline requirements or strategic reserve obligations for the components whose destruction would take those facilities offline for a year or more. The gap between “we must protect this facility” and “we must be able to recover this facility within a defined timeline” is where the regulatory framework currently ends.

For the generation investor, this gap means that physical damage to a main power transformer — from any cause — produces an outage of eighteen to twenty-four months with no regulatory backstop requiring faster recovery. The risk is shared across all transmission-connected generation, not specific to any technology. But it is a risk that current regulations do not require anyone to mitigate.

The pathway to closing this gap runs through an extension of NERC CIP-014 to include recovery timeline requirements, and through the Department of Energy’s existing authority under the Federal Power Act to address strategic equipment reserves. Neither step requires new legislation. Neither has been taken.

Part Five: Where the Analysis Goes Next

The framework described in this document and developed in full in the companion technical white paper is a foundation, not a finished structure. Two extensions are already identified as the next analytical work, and both have direct implications for the investment and policy communities.

The Grid Stability Connection

There is a physical property of synchronous generators — nuclear, coal, gas turbines, and large hydro — that does not appear in any cost comparison but has significant consequences for grid stability. When a synchronous generator is running, the spinning mass of its turbine and generator stores kinetic energy. If another generator on the grid trips suddenly and the grid loses power, every synchronous machine in the interconnection automatically and instantaneously begins exchanging that stored energy with the grid — slowing down slightly to inject power into the system and arrest the frequency decline. This happens before any control system acts. It is physics, not software.

Wind turbines, solar panels, and battery storage systems do not do this. They connect to the grid through power electronics and do not store rotational energy. Advanced inverter configurations can simulate this inertial response through software — so-called synthetic inertia —, but this capability is not standard in currently deployed systems, depends on available stored energy and functioning control software, and has different physical characteristics from the mechanical inertia of a spinning turbine. For the current North American grid, the practical inertia contribution of inverter-based resources remains close to zero. As the generation mix shifts toward these sources, the grid’s total stored rotational energy — its inertia — declines. A grid with lower inertia responds more violently to the sudden loss of a large generator: frequency drops faster, drops further, and is more likely to trigger automatic load shedding before it recovers.

There is a further complication that the investment and policy communities have not fully absorbed. Even where synthetic inertia from advanced inverters is available, the devices providing it are manufactured predominantly in China. Huawei and Sungrow alone hold approximately forty to fifty percent of the global utility-scale inverter market share. The broader IBR supply chain — solar panels, wind turbine components, rare earth magnets — follows the same pattern of Chinese dominance. The control algorithms that implement synthetic inertia are proprietary firmware running on those devices.

This creates a risk category that sits between supply chain disruption and cybersecurity. CISA and the Department of Energy have both raised specific concerns about Chinese-manufactured inverters in the United States grid. Documented cases of undisclosed remote access functionality — communication capabilities not listed in product specifications — have been found in inverters from Chinese manufacturers already operating in the field. A grid-forming inverter that withdraws its synthetic inertia response at a moment of grid stress, or that provides a destabilizing rather than stabilizing response, would be indistinguishable from normal inverter behavior until the moment of consequence. The attack surface is embedded in the generation fleet, not external to it.

Physical inertia from a synchronous generator cannot be hacked. It is a rotating mass governed by physics. Synthetic inertia from a Chinese-manufactured inverter running proprietary firmware is a different proposition entirely. The technical white paper addresses this distinction in Section 5.6. For investors, the practical implication is that a generation portfolio relying on synthetic inertia from IBRs as a grid stability backstop is relying on a mechanism whose reliability is not purely a technical question — it is also a geopolitical and supply chain question that no current regulatory framework fully addresses.

Load shedding is, from a data center’s perspective, an outage. It does not matter that the outage was caused by an event somewhere else on the grid — the data center loses power just as surely as if its own generation source had failed.

The next stage of the analytical work connects the generation mix composition directly to the probability of these load shedding events, through the physical equations that govern frequency response. The result will be a quantitative estimate of how much the risk of grid-triggered outages increases as the mix shifts away from synchronous generation. This risk is currently invisible to investors and unpriced in any market.

The Targeting Analysis

The parametric framework that reveals which generation sources are most valuable to the grid — highest productive utilization, longest fuel independence, highest rotational inertia — also, by straightforward logic, reveals which sources an adversary seeking to maximize grid disruption would target first. The technical white paper develops this as the “adversarial dual” of the Pareto frontier optimization (see Note 1).

The next stage of the analytical work develops this mirror analysis formally: given a generation portfolio, which unit, if destroyed, produces the largest and longest consequence for the grid and for the loads that depend on it? The answer follows from the same parameters that define investment value — the source with the highest productive utilization, the longest fuel independence, and the highest inertia contribution is simultaneously the highest-value grid asset and the highest-value target.

For infrastructure protection planning, this analysis produces a targeting priority ranking that is derived from the physics and economics of the grid rather than asserted from general principles. For investors, it provides a quantitative basis for assessing which of their generation assets face the highest physical risk, and what the consequences of losing those assets would be for the contracts they underpin.

Both extensions are identified in the companion technical white paper as forward work. They are noted here because their results will be directly relevant to the investment and policy communities that this document addresses.

The Single Finding

This document has covered a lot of ground—a new kind of electricity customer. Three things the standard scorecard misses. The real cost of firming variable sources to continuous output. The fuel and equipment questions that no current comparison tool addresses. Three regulatory gaps that leave the grid less protected than investors may assume. Two analytical extensions that will sharpen the picture further.

All of it converges on one finding.

For the specific problem of serving continuous, large-scale, low-carbon, resilient electricity demand — which is exactly the problem the hyperscale data center surge has placed before the investment community — nuclear is the only available generation source that satisfies all of the requirements simultaneously. Not because of any assumption favorable to nuclear power built into the analysis. Because of the arithmetic.

This finding does not resolve the permitting challenges, the construction timelines, or the social and political constraints on nuclear development in various jurisdictions. Those are real, and investors will need to navigate them. What the finding does establish is what those constraints cost: a portfolio that excludes nuclear must accept either higher carbon intensity, or higher electricity cost, or lower grid resilience, or some combination of all three. The framework makes those tradeoffs visible. The judgment about whether they are acceptable belongs to the decision makers, not to the analysis.

The investment community is being asked to commit very large amounts of capital to generating assets that will operate for decades. Making that commitment with the right scorecard — one that asks what continuously delivered, low-carbon, resilient power actually costs rather than what nameplate capacity costs on average — is not a minor refinement. It is the difference between buying the right tool for the job and discovering, years later and at great expense, that the tool that looked cheapest on the price tag was the wrong one all along.

Note

Note 1 — The Pareto Frontier

The term “Pareto frontier” comes from optimization theory and is easier to understand through an example than through a definition.

Suppose you are buying a car and you care about two things: fuel economy and horsepower. You want as much of both as possible. But they are in tension — more horsepower generally means worse fuel economy, and vice versa. Now imagine plotting every available car on a graph. Fuel economy on one axis, horsepower on the other. Some cars are simply bad on both measures — low horsepower and poor fuel economy. You would never choose them, because other cars beat them on both dimensions simultaneously. These are dominated options.

But some cars represent the best possible tradeoff available at their particular balance point. If you want more horsepower than that car offers, you have to accept worse fuel economy. If you want better fuel economy, you have to give up horsepower. No car gives you more of both simultaneously. These cars form the Pareto frontier — the boundary along which every improvement in one objective requires a sacrifice in the other.

In the companion technical white paper, the two objectives are minimizing carbon intensity and maximizing resilience. The Pareto frontier is the set of generation portfolio compositions where you cannot improve the carbon outcome without worsening the resilience outcome, and vice versa. Two results of particular interest sit off the frontier in different directions.

Sources in the dominated region — CCGT and coal — are worse than available alternatives on both objectives simultaneously. No well-constructed portfolio chooses them for long-term building.

Sources that are Pareto-dominant — nuclear — improve both objectives simultaneously when added to the portfolio. There is no tradeoff when moving toward nuclear; both carbon intensity and resilience get better together. The optimization pushes nuclear to its maximum feasible allocation before the tradeoff frontier is encountered at all.

Wind and solar live on the frontier itself: they improve the carbon objective while worsening the resilience objective through inertia reduction. That is a genuine tradeoff — neither dominated nor dominant — and the Pareto frontier is where that tradeoff is made explicit.

The adversarial dual referenced in Part Five is the mirror image of this optimization: rather than asking which portfolio composition best satisfies both objectives, it asks which unit an adversary should target to cause the most damage to both simultaneously. The unit that scores highest on productive utilization, fuel independence, and inertia contribution is the answer to both questions — most valuable to defend, most valuable to destroy.

Reference

Roxey, T. (2026). “Generation Mix Optimization Under Competing Objectives: A Parametric Framework for Climate and Resilience Tradeoffs.” Eclectic Technologies White Paper, Version 0.3.

Additional data sources underlying the parameter values discussed in this document are cited in full in the companion technical white paper, including: EIA Electric Power Annual 2024; NERC 2025 Long-Term Reliability Assessment; Lazard Levelized Cost of Energy Analysis v17; IPCC AR6 lifecycle emissions ranges; and IEEE Std 1110 for synchronous machine inertia characteristics. Fuel independence duration and recovery timeline values are analytically estimated by the author from engineering first principles, as described in the technical white paper.